Submission preview

Setup
Assembly file: SPAdes on dataset 15 and 16 and collection 12: Contigs
Local folder: /Users/josesanchez/Desktop/Bioinformatics/BIOL4810/module02/Annotation

Prokka run
Version: Galaxy Version 1.14.6+galaxy1
Settings: Force GenBank Compliance -- YES

Outputs understanding
.gff vs .ffn vs .faa: .gff is a file that lists the information/annotations about the genes such as names of the genes, their function, and proteins (known or potential). .ffn files provide the nucleotide sequences for the genes in the genome. .faa files provide the amino acid sequences for the proteins encoded by the genome.

Prokka summary (.txt)
Organism: organism: Genus species strain
Contigs: contigs: 79 *Bases:* bases: 4797060
CDS: CDS: 4507 *rRNA:* rRNA: 3 *tRNA:* tRNA: 62 *tmRNA:* tmRNA: 1
Sanity check: Yes, the genome size and CDS count does make this appear as though it is a typical bacterial genome. This is because the genome size is between 4 - 6 million bases and the number of CDS is between 3,500-5,500 coding sequences.

rRNA evidence
16S description: gnl|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - . ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal RNA
23S description: gnl|Prokka|NCJPFIPA_53 barrnap:0.9 rRNA 4 1724 0 + . ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)
5S description: gnl|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + . ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal RNA

CheckM2
CheckM2 version: Galaxy Version 1.1.0+galaxy0
Completeness: 98.96%% *Contamination:* 0.1%%
CheckM2 interpretation: I would trust this assembly for annotation and species identification. One reason is the low contamination rate, indicating that the reads are mainly from the intended organism of interest. Additionally, the high percentage of completeness indicates that the assembly contains a majority of the genetic information.

Hypothetical proteins
How estimated: To find the number of hypothetical proteins, I used Command + f and searched for "hypothetical protein".

 # hypothetical: 1,098 *% hypothetical:* 24.36%
Interpretation: A high percentage of hypothetical proteins means that there are many portions of the genome that are predicted to code for a protein, however, it is not known for certain. A high percentage of hypothetical proteins does not mean that the protein does not serve a purpose or function.

16S extraction
16S header: >NCJPFIPA_04563 16S ribosomal RNA
16S sequence:

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TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT
GTGAAATCCCCGGGCTCAACCTGGGAACTGCATTCGAAACTGGCAGGCTTGAGTCTTGTA
GAGGGGGGTAGAATTCCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGT
GGCGAAGGCGGCCCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAAC
AGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCTACTTGGAGGTTGTGCCCTT
GAGGCGTGGCTTCCGGAGCTAACGCGTTAAGTAGACCGCCTGGGGAGTACGGCCGCAAGG
TTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCG
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
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Why 16S: We use 16S for quick ID because this gene is generally conserved among prokaryotes.

16S BLAST
Top hit: Salmonella enterica subsp. enterica serovar Typhimurium strain SAP17-7299 chromosome, complete genome
Metrics: % identity - 100%; Query Coverage - 100%
Interpretation: Based on the results, the organism is likely to be Salmonella enterica. This is due to 100% of the sequence that was entered into BLAST aligning with the sequence in the database. Additionally, the 100% percent identity represents the idea that the two sequences at a match and the E value of 0 means that the match did not occur due to chance.
Limitation: 16S BLAST can be misleading because this sequence is generally conserved among prokaryotes. This means that the 16S sequence could be very similar if not the same between two strains or two closely related species.

BLAST screenshot

TYGS

TYGS vs 16S: Compared to 16S BLAST, TYGS is going to look at the entire genome whereas 16S BLAST looked at solely the gene encoding the 16S rRNA. Also, TYGS will use relationships and the entire genomes to establish a phylogenetic tree. In doing so, it is easier to establish or distinguish between the relationships of the selected organisms because there is more information available to compare rather than just the 16S rRNA.
Closest match + dDDH: Salmonella typhimurium JCM 1652; dDDH: 99.2
70% threshold: The 70% dDDH threshold means that if the genomes are 70% similar or more, then they are considered to be members of the same species. Anything below this threshold are considered different species.
Proteome tree option (low support): If TYGS has low branch support, then that means that there is less confidence in the relationships that are shown, however, there can be greater confidence if a proteome-based tree is offered. In using a proteome-based tree, phylogenetic signal and support can be improved as well-conserved protein sequences (or amino acids) can help minimize any noise that would be present if we were looking at the gene sequence itself.

iTOL tree

Tree adds value because: This tree tells me about the relatedness of the species listed and the evolutionary relationships among them. It also provides the branch support values for each node/relationship, revealing the confidence in the individual relationships.
Support/trust: Branches that I would trust is the placement of the Salmonella typhimurium (Pine Siskin) along with the other S. typhimurium stains. This is due to the higher support values near the nodes (ex. 100, 91, or 97). A branch that I would be more cautious about are the ones with the lower support values such as the one with a value of 47. This is because having a lower support value indicates that there is not as much confidence in these relationships when the tree was reconstructed multiple times.

iTOL tree screenshot

Workflow note (for Git)

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Step 1) Filtering contigs shorter than 500 bp ; Tool: Filter FASTA ; Version: Galaxy
Version 2.3 ; Parameters: Filtered on sequence length, minimum length 500 bp
Step 2) Quality assessment using CheckM2 ; Tool: CheckM2 ; Version: Galaxy Version
1.10+galaxy0 ; Parameters: Model Options - Output quality prediction for both models
for each genome
Step 3) Genome annotation ; Tool: Prokka ; Version: Galaxy Version 1.14.6+galaxy1 ;
Parameters: Force GenBank compliance - YES
Step 4) Species ID ; Tool: BLAST ; Version: BLASTn (web) ; Parameters: rRNA/ITS
database
Step 5) Build phylogenetic tree ; Tool: Type Strain Genome Server ; Version:
https://tygs.dsmz.de/ (web) ; Parameters: Whole Proteome Tree
Step 6) Visualize phylogenetic tree ; Tool: Interactive Tree of Life ; Version: v7
(web) ; Parameters: Display bootstraps as text
Step 7) Average nucleotide identity ; Tool: https://www.ezbiocloud.net/tools/ani ;
Version: Website ; Parameters: From NCBI Genome Database
(https://www.ncbi.nlm.nih.gov/datasets/genome/), downloaded the FASTA file for
Salmonella Typhimurium (closest hit on 16S BLAST) and Salmonella bongori (more
distant relative). Compared FilterContigs.fasta to the FASTA files from S.
Typhimurium and S. bongori.
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Name	Last commit message	Last commit date
..		
.DS_Store	uploading annotation folder from bioinformatics	1 hour ago
AnnotationNotes.rtf	uploading annotation folder from bioinformatics	1 hour ago
BLASTResultScreenshot.png	uploading annotation folder from bioinformatics	1 hour ago
CheckM2Report.tabular	uploading annotation folder from bioinformatics	1 hour ago
FilterContigs.fasta	uploading annotation folder from bioinformatics	1 hour ago
Prokka.faa.fasta	uploading annotation folder from bioinformatics	1 hour ago
Prokka.ffn.fasta	uploading annotation folder from bioinformatics	1 hour ago
Prokka.genbank	uploading annotation folder from bioinformatics	1 hour ago
Prokka.gff3	uploading annotation folder from bioinformatics	1 hour ago
Prokka.txt	uploading annotation folder from bioinformatics	1 hour ago
SBongori.fna	uploading annotation folder from bioinformatics	1 hour ago
STyphimurium.fna	uploading annotation folder from bioinformatics	1 hour ago
SalmonellaTree.pdf	uploading annotation folder from bioinformatics	1 hour ago
SalmonellaTree.phy	uploading annotation folder from bioinformatics	1 hour ago